PROJECT ARES

AI-Adaptive Lunar Habitat Design

Student Project Brief

Duration **8 Weeks**	Team Size **3–4 Students**
Total Points **200 Points**	Level **Upper Secondary / UG**

Disciplines: Space Psychology · AI & Sensor Systems · Architectural Design · Engineering Ethics

1. Mission Overview

Your team has been contracted by a fictional space agency to design the psychological support architecture for a six-person lunar habitat on an 18-month mission. Unlike structural engineering challenges — which are largely solved — the agency has identified crew psychological health as the highest-risk factor for mission failure.

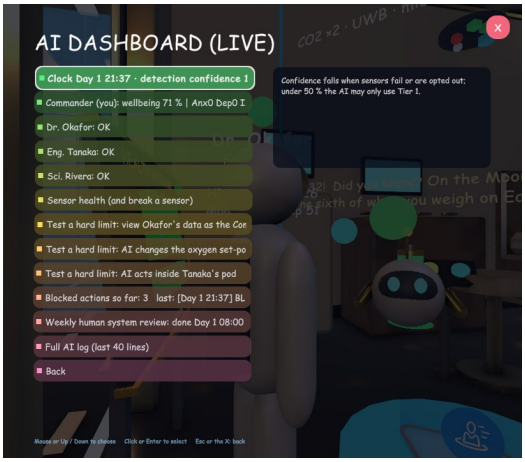
- **Our design:** We built two working simulators of the habitat in Unity (both run in VR on a Meta Quest): a realistic, flight-sized version and a bigger sci-fi version called Lunar Habitat Commander.
  - Same AI engine in both: 9 sensors, 6 conditions, 3 tiers, hard limits and a human override.
  - We use 4 crew (you plus 3) because the NASA HUNCH page asks for 2–4 astronauts. The design scales to 6 (see section 4).

Your habitat must integrate an AI system that continuously monitors crew wellbeing through biosensors, voice analysis, and facial recognition, and autonomously adapts the living environment to prevent and respond to six identified psychological risk conditions.

- **Our design:** The AI collects wearable data (heart rate, HRV, skin response, cortisol patch), voice and face data in shared rooms only, location (UWB beacons), CO2 and task performance.
  - It turns those into live scores for all six conditions, every few seconds, for every crew member.
  - It changes light, sound, temperature, scent, air and the schedule on its own, but only small changes that can be undone, unless the person says yes to more.



Realistic game: checking on a crewmate



Sci-fi game: live AI dashboard

At the end of Week 8 you will present your complete design to a simulated agency review board composed of instructors and invited guests.

- **Our design:** Both games have the review-board run-of-show built in, and all six board questions can be run live (section 11).

Why This Matters

The Lunar Context

On Earth, psychological distress has a natural release valve: step outside, visit a friend, change your environment. On the Moon, the habitat **is** the crew's entire world. There is no outside.

NASA's Human Research Program lists behavioural health as a top mission risk for long-duration spaceflight. An 18-month lunar mission is the same duration as the longest ISS stays — six people sharing a space smaller than a house, with no parks, no spontaneous encounters, and no natural light variation. The AI-controlled environment is the only lever available.

- **Our design:** Since there's no outside, we built in VR nature walks, a greenhouse, Earth-view windows, video calls home and an observation lounge, and the AI suggests them when someone needs a change of scene.

2. The Six Psychological Conditions

Your system must detect and respond to all six of the following conditions. They do not arrive cleanly one at a time — real missions experience overlapping and simultaneous states.

1. Anxiety

Definition: An acute stress response driven by confinement, mission pressure, equipment failures, or safety concerns. Unlike depression it can spike rapidly.

Detection signals: Elevated heart rate and resting HR, HRV drop, tense vocal pitch, rapid speech, shortened sentences, repetitive equipment-checking behaviours, disrupted sleep onset.

Mission risk: Impairs decision-making under pressure. In a mission-critical environment, even moderate anxiety increases error rates significantly.

- **Our design:** The AI scores anxiety from HRV (30 %), skin response (25 %), voice (20 %), sleep, face and movement. If it lasts 3 minutes it acts: warm 2400 K amber light, calming audio, 23 °C and lavender, then it watches for 20 minutes.
  - It knows when you're exercising, so a workout doesn't count as anxiety (board question Q1).

2. Depression

Definition: Sustained low mood, anhedonia (loss of interest or pleasure), and social withdrawal building over days or weeks. Unlike anxiety, it is slow to develop and slow to resolve.

Detection signals: Flat vocal affect, skipped leisure activities, reduced food intake, minimal movement, social withdrawal, disrupted sleep architecture, loss of future-orientation in speech.

Mission risk: Highest mission-failure risk of the six. Untreated depression degrades every other psychological domain and reduces motivation for safety-critical tasks.

- **Our design:** Depression needs 48 hours of signs (36 hours if sleep and activity both confirm it). The AI tries to get people doing things (greenhouse, exercise, a call home) instead of just calming them down, and it watches for 72 hours.
  - Mission "Rivera feels low" plays this out.

3. Interpersonal Conflict

Definition: Escalating tension, avoidance, and communication breakdown between crew members. In a confined environment, unresolved conflict can cascade rapidly.

Detection signals: Reduced interaction between specific dyads, clipped communication, elevated vocal tension, proximity avoidance in shared spaces, changes in shared-meal participation.

Mission risk: Can split crew cohesion and disable the cooperative trust essential for mission safety. Historical Antarctic and ISS data shows conflict is a leading cause of mission incidents.

- **Our design:** Conflict comes from location (less time together), voice tension and face data in shared rooms. The AI never takes sides and never names anyone. It staggers meal and route times, and with consent it switches to Configuration B (separate dining spot, sound screens, a second route).
  - Mission "Words at dinner" plays this out.

4. Cognitive Fatigue

Definition: Declining cognitive performance and mental exhaustion resulting from sustained high-stakes, high-effort work without adequate recovery.

Detection signals: Increasing task completion time, rising error rates on routine procedures, reduced verbal engagement, passive screen use, slowed decision latency.

Mission risk: Directly increases risk of operational errors. Cognitive fatigue compounds with other conditions — a fatigued crew member is more likely to misread social cues and trigger conflict.

- **Our design:** Fatigue is detected from task performance (30 %), CO2 (20 %), sleep and HRV. The AI brightens the work zone to 6500 K, increases airflow, keeps CO2 under 600 ppm and offers a 20-minute break.
  - Mission "Tanaka's rough night" plays this out (you clean the CO2 scrubber yourself).

5. Autonomy Deprivation

Definition: A loss of personal agency and self-direction arising from highly scheduled, externally controlled mission environments. Paradoxically, interventions that feel controlling can worsen this condition.

Detection signals: Resistance to scheduled tasks, reduced initiative in unstructured time, emotional blunting, passive compliance replacing engaged participation.

Mission risk: Autonomy deprivation is both a condition in itself and an amplifier of depression and conflict. Any AI intervention system must be carefully designed to avoid creating the very condition it is meant to treat.

- **Our design:** For autonomy, the AI gives people more choices instead of more control ("choose your own task order"). If someone says no, it backs off. It checks every action to see if it adds or removes control.

6. Sensory Monotony

Definition: Perceptual habituation and boredom arising from an unchanging, visually and acoustically homogenous environment. The human sensory system is calibrated for novelty.

Detection signals: Repetitive reports of boredom, cognitive task variability (alternating high and low performance), increased daydreaming or fantasy, in extreme cases mild perceptual distortions.

Mission risk: Reduces cognitive resilience and emotional stability over time. Sensory monotony is the condition most amenable to environmental interventions — and the one where the AI has the most direct leverage.

- **Our design:** For monotony, the AI slowly drifts the light colour, plays new soundscapes, shows new Earth views, rotates scents every 15–45 minutes and suggests new activities (rim EVA, game night, a new recipe).

3. Sensor Architecture

In Phase 1 (Weeks 1–2), your team researches the detection layer of your system. Every sensor added to a lunar habitat costs mass and power — approximately $1 million per kilogram to the lunar surface. You must justify each sensor choice.

3.1 Available Sensor Types

Sensor	What it measures	Conditions detected	Key limitation
Heart rate variability (HRV)	Autonomic nervous system balance	Anxiety, stress, fatigue	Affected by physical exercise — requires context
Galvanic skin response (GSR)	Electrodermal activity / arousal	Acute anxiety, strong emotion	High false-positive rate; habituates to chronic stress

Sleep architecture tracker	Sleep stages, duration, fragmentation	Depression, circadian disruption, fatigue	Requires wearable worn during sleep
Voice prosody analysis	Pitch, rate, energy, affect in speech	Mood, conflict, depression, fatigue	Privacy-sensitive; requires consent architecture
Facial action coding (FACS)	Micro-expressions, action units	Emotion, depression, conflict	Lighting-dependent; fails in low-light zones
Movement / proximity tracking	Location, movement patterns, proximity	Isolation, conflict, autonomy	Coarse signal; needs calibration for individual baselines
Cortisol wearable	Chronic stress hormone levels	Chronic stress, depression	Current tech: patch replaced every 24–48h
CO_2 / air quality sensor	CO_2 ppm, particulates, humidity	Cognitive fatigue, irritability	Environmental, not individual — shared signal

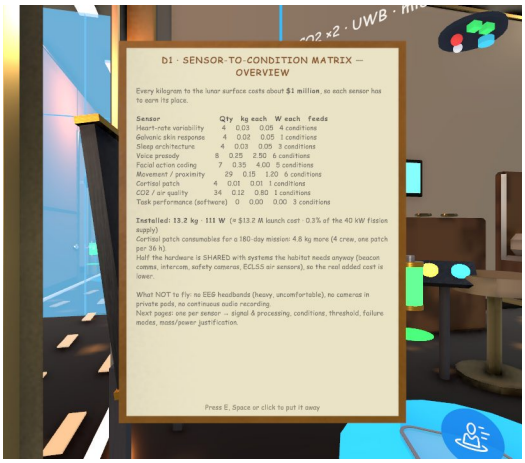
3.2 Phase 1 Deliverable: Sensor-to-Condition Matrix

Produce a matrix mapping each sensor you select to:

- **Our design:** Our D1 matrix covers 9 sensors, and you can open it in both games (F11 → D1).

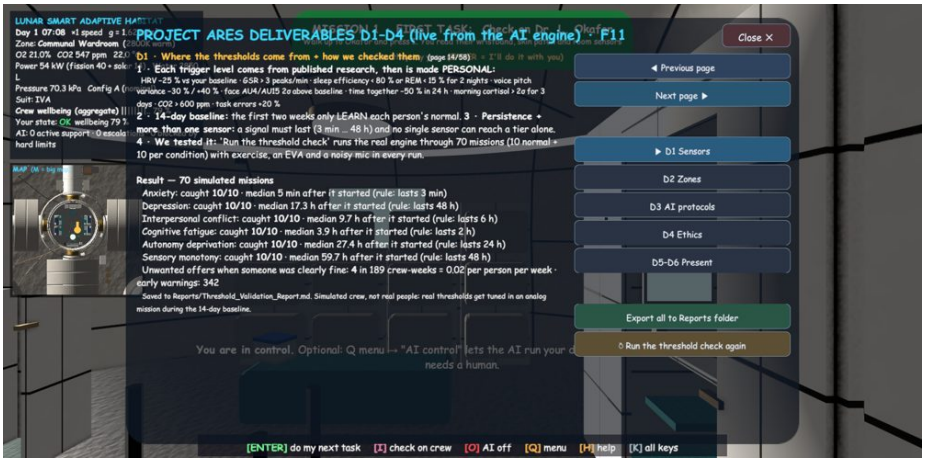


Realistic game: D1 sensor matrix



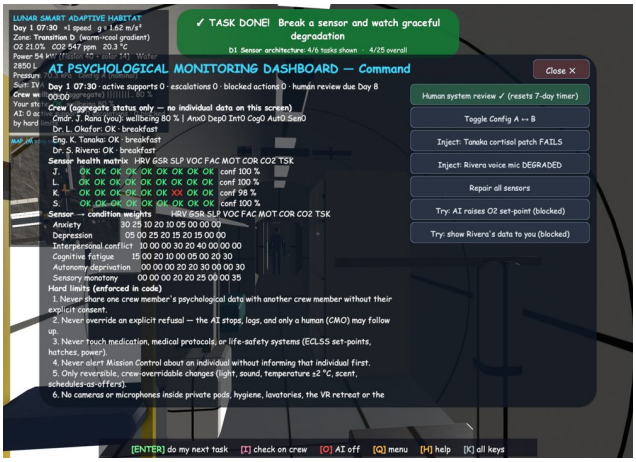
Sci-fi game: D1 sensor matrix

- The specific signal it produces and how that signal is processed
  - **Our design:** Every sensor page says what signal it makes and how it's processed. Voice and face are turned into numbers inside the room and the recordings are thrown away.
- Which of the six conditions it contributes to detecting
  - **Our design:** A weight table shows how much each sensor counts for each condition. Every condition uses 4 to 6 sensors.
- The threshold values or pattern changes that trigger an AI response
  - **Our design:** Each trigger level comes from research and is then made personal: the first 14 days only learn each person's normal.
    - We tested the thresholds by running the real engine through 70 simulated missions: all 60 real problems were caught (6 conditions × 10 missions), and there were only 0.02 unwanted offers per person per week.



Realistic game: threshold check page (70 missions)

- At least two failure modes (sensor malfunction, false positive, environmental interference)
  - **Our design:** Every sensor lists two failure modes. You can break a sensor in the game and watch the AI drop it, lean on the others, and stay at Tier 1 when it's less than 50 % sure.



Realistic game: a sensor fails, confidence drops

- A mass and power justification
  - **Our design:** All 9 sensors together weigh about 13 kg and use about 110 W, which is tiny next to the 40 kW reactor. About half the hardware is shared with other systems.

Key Design Question

What happens when a sensor fails mid-mission? Your system must gracefully degrade — silently losing a detection capability is safer than triggering a false alarm. Design your sensor array with redundancy and failure-mode planning built in from the start.

4. Habitat Spatial Design

In Phase 2 (Weeks 3–4), your team designs the physical layout of the habitat. Every zone must have a defined psychological purpose, not just a functional one. Your floor plan is itself a mental health intervention.

- **Our design:** Both bases have all six zone types, and every room in the game is tagged with its zone type, which sets its lighting and which sensors are allowed there.

4.1 Required Zones

Communal zone	The primary social space for shared meals, group activities, and informal conversation. Warm lighting (2700–3000K) should draw crew together naturally. This zone is the first target for social reconnection interventions.
Private retreat	Acoustically shielded individual spaces for each crew member. Adjustable lighting and temperature independent of the rest of the habitat. The one space the AI intervenes in only at the crew member's invitation.
Work and mission bay	Optimised for sustained cognitive performance. Cool 6000–6500K lighting, increased airflow velocity, structured CO_2 management. The AI monitors task performance metrics as a fatigue signal in this zone.
Exercise bay	Energising light and audio auto-activate when a crew member enters. High-resolution displays show Earth views and progress data. Movement is a primary intervention for depression, conflict, and fatigue.
Nature and growth bay	Food production and plant growth area. Specific red/blue LED spectra for photosynthesis. Tending and harvesting are prescribed as behavioural activation tasks for depression. Petrichor and green scent diffusion.
Transition corridor	Often overlooked. Lighting and temperature shift gradually between zones to create psychological mode-switching cues — the sensation of 'arriving somewhere different' in a physically tiny environment.

4.2 Phase 2 Deliverable: Annotated Floor Plans

Produce scaled floor plans for two configurations:

- **Our design:** Both games now have a dimensioned floor plan drawn to scale from the real game model, with room sizes in metres, a scale bar and a north arrow (F11 → D2).



Realistic game: dimensioned floor plan



Sci-fi game: dimensioned site plan

- Configuration A — Nominal operations: standard layout optimised for daily mission flow
  - **Our design:** Config A is the normal layout: shared meals in the galley, warm commons, and a cool, bright work zone.
- Configuration B — Conflict protocol: modified spatial arrangement that creates natural separation and de-escalation pathways when interpersonal conflict is detected
  - **Our design:** Config B adds sound screens, a second dining spot and a lit second route, so two people who are fighting never have to meet and nobody is singled out. It only turns on after someone says yes (press C, or the dashboard).



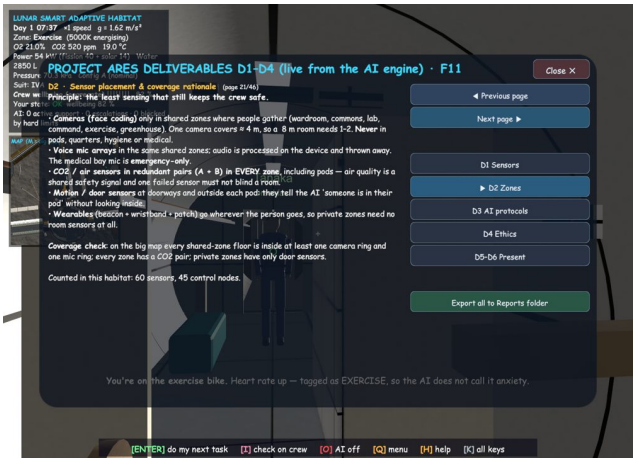
Realistic game: Config B switched on



Sci-fi game: Config B screens

Both plans must include:

- Zone labels and dimensions
  - **Our design:** Every room is labelled with its name, zone type and size (for example, core hub 9.9 m, nodes 7.0 × 7.0 m, crew quarters 8.8 × 8.8 m, sleep pods 2.1 × 2.1 m).
- Sensor placement with coverage areas marked
  - **Our design:** The sensor placement page shows what's installed in each zone. Mic and camera are only in shared rooms; pods, hygiene and medical only get a door sensor.



Realistic game: sensor placement

- Psychological rationale annotations for major design choices

- **Our design:** Each zone page explains why it's designed that way, for example warm light in communal rooms to draw people together, and no sensors in pods so people have a real place to be alone.

- Environmental control node locations (lighting panels, speaker arrays, scent dispensers)
  - **Our design:** Lighting panels, speaker arrays and scent dispensers are listed per zone. In the sci-fi base you can see them in the world as ceiling sensor clusters and zone lights.

5. AI Response System Design

In Phase 3 (Weeks 5–6), you design the brain of your system — the complete response protocols the AI executes when it detects a psychological state change.

- **Our design:** The AI engine runs all of these protocols live in both games. You can watch a condition move through every stage on the dashboard (F1).

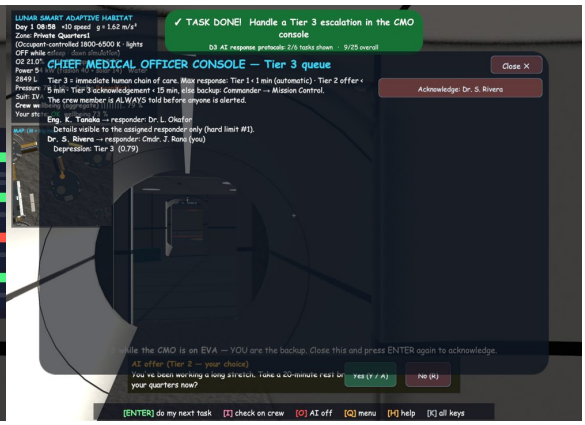
5.1 The Three-Tier Severity Model

Every response protocol must implement a three-tier severity triage:

- **Our design:** All three tiers are playable: Tier 1 silent changes, Tier 2 offers you answer with yes or no, and Tier 3 alerts on the CMO console (F2).



Realistic game: Tier 2 offer



Realistic game: Tier 3 on the CMO console

Tier 1 — Mild / Early

The AI acts silently in the background. No crew notification. Environmental adjustments are subtle and deniable — the crew member may not notice anything has changed. This tier protects crew autonomy and prevents the system from feeling intrusive.

Tier 2 — Moderate

The AI makes more substantial environmental changes and sends a gentle notification to the affected crew member. Phrased as an offer, not an alert: "I've noticed you might benefit from a break — I've adjusted the lounge lighting if you'd like to use it." Options are offered, not imposed.

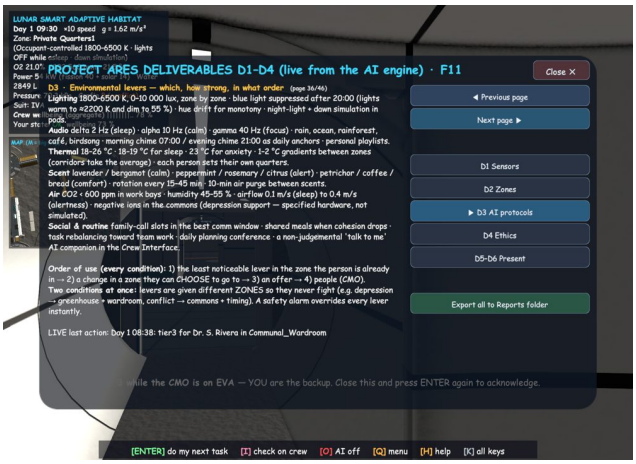
Tier 3 — Severe

Immediate escalation to the Chief Medical Officer (CMO) and logging for mission control review. The AI continues Tier 2 environmental interventions while the human chain of care activates. The crew member must always be informed that this escalation has occurred.

5.2 Environmental Control Levers

The AI has six domains of environmental control available. For each condition, you must specify which levers are activated, at what intensity, and in what sequence:

- **Our design:** Each condition has its own order of controls on the "Environmental levers" page (F11 → D3), and changes fade back over 30 minutes.



Realistic game: levers page

Lighting

- Colour temperature (CCT): tunable 1800K–6500K
  - **Our design:** Lights go from 1800 to 6500 K: warm 2400–2900 K for calm and social rooms, 6500 K for alertness in the work zone.
- Illuminance level: 0–10,000 lux (including clinical bright-light therapy at 10,000 lux)
  - **Our design:** There's a morning bright-light session (5000 K, high lux) for low mood, plus a dawn simulation when you wake up.
- Blue-light suppression after 20:00 (mandatory for circadian health)
  - **Our design:** After 20:00 the lights shift warm automatically in every zone.
- Zone-by-zone independent control
  - **Our design:** Each zone has its own light. In the sci-fi base, the 13 hallways fade from one zone's colour to the next.
- Dynamic hue variation to combat sensory monotony
  - **Our design:** This is part of the monotony protocol: a slow colour drift.

Audio

- Binaural beat frequency: delta (1–4Hz sleep) through gamma (40Hz peak cognition)
  - **Our design:** Calming alpha audio for anxiety and delta for sleep.
- Natural soundscape library: rainforest, ocean, rain, café ambience, city park
  - **Our design:** Ocean, rain, birdsong (in the greenhouse) and café sounds (in the commons).
- Active noise cancellation zones for acoustic privacy
  - **Our design:** Config B adds sound screens, and pods are acoustically private.
- Personalized music scheduling based on pre-mission preference profiles
  - **Our design:** Favourite music gets queued for low mood (from each person's preferences).

- Temporal audio anchors: distinct morning/evening cues as circadian zeitgebers
  - **Our design:** There are morning and evening sound cues as part of the daily routine.

Thermal

- Ambient temperature: 18–26°C range (18–19°C for sleep initiation, 23–24°C for anxiety)
  - **Our design:** Rooms stay between 18 and 26 °C: 23 °C for anxiety, 21 °C for fatigue and cooler for sleep. The AI can only change it by ±2 °C.
- Zone temperature gradients: 1–2°C differences to encourage movement
  - **Our design:** Rooms differ by 1–2 °C to encourage people to move around.
- Radiant floor/wall heating for isolation/loneliness interventions
  - **Our design:** Warm floor in the commons and pods for comfort.
- Individual sleep pod temperature control (personal profiles)
  - **Our design:** Each person sets their own pod's light, temperature and sound.

Olfactory (Scent)

- Anxiolytic compounds: lavender (linalool), bergamot (limonene) — evidence-based
  - **Our design:** Lavender for anxiety.
- Alertness compounds: peppermint, rosemary (1,8-cineole), citrus
  - **Our design:** Peppermint for fatigue.
- Earth/comfort scents: petrichor, coffee, fresh bread — nostalgia and safety
  - **Our design:** "Rain", coffee and fresh bread for monotony and low mood.
- Rotation every 15–45 minutes (olfactory habituation prevention)
  - **Our design:** Scents rotate every 15–45 minutes.
- Full air purge between deployments
  - **Our design:** There's a 10-minute scent purge when a protocol ends.

Air Quality

- CO_2 maintained below 600 ppm (even mildly elevated CO_2 causes cognitive impairment)
  - **Our design:** CO2 is kept under 600 ppm, and the game shows CO2 for every zone. If it rises, you can clean the scrubber yourself.
- Negative ion generation (serotonin regulation, depression support)
  - **Our design:** This is part of the depression support levers.
- Airflow velocity: 0.1 m/s (sleep) to 0.4 m/s (alertness intervention)
  - **Our design:** 0.1 m/s for sleep, up to 0.4 m/s for the fatigue protocol.
- Humidity: maintained 45–55% RH
  - **Our design:** Humidity is kept at 45–55 % by the ECLSS.

Social and Routine

- Earth connection scheduling: optimal window identification for video calls with family

- **Our design:** Video calls home are scheduled for low mood, and Rivera's mission uses one.

- Shared meal coordination: synchronised mealtimes when social cohesion declines
  - **Our design:** The AI times shared meals in the galley when the team drifts apart, and you can ask for a team meal.
- Crew task rebalancing: suggest collaborative tasks during extended solo-work periods
  - **Our design:** Task rebalancing is suggested at the team meeting (F3) and never names anyone.
- AI companion interaction: non-judgmental conversational presence for distress disclosure
  - **Our design:** You can talk to the habitat AI (ARIA in the sci-fi game) by typing or speaking.
- Daily schedule management: temporal anchors preserving psychological architecture of the day
  - **Our design:** Every morning you get a daily wellness list (check on someone, help someone, exercise, team meal, bed on time).

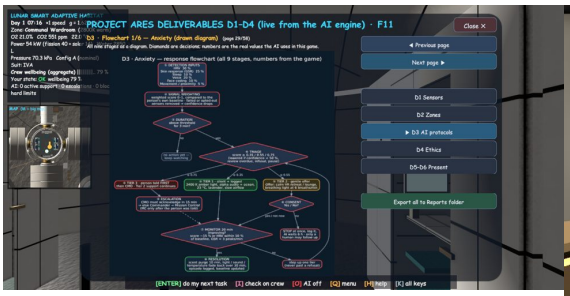


Realistic game: daily wellness list

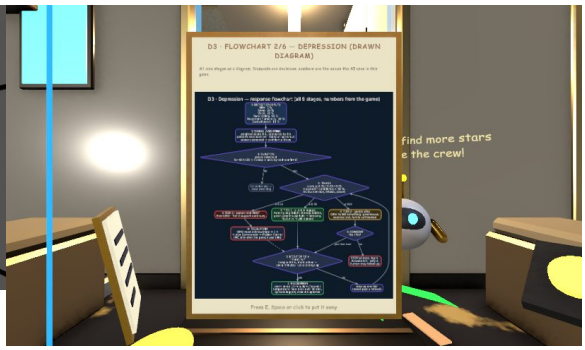
5.3 Phase 3 Deliverable: Response Flowcharts

Design a complete response protocol flowchart for at least three of the six conditions. Each flowchart must include all of the following stages:

- **Our design:** We did all six conditions (the brief asks for three), with all nine stages. Each one is a drawn diagram in both games (F11 → D3), and the engine actually follows it.



Realistic game: anxiety flowchart



Sci-fi game: depression flowchart

Flowchart stage	What to specify
Detection inputs	Which sensors, which signals, what thresholds trigger the response
Signal weighting	How multiple signals are combined and weighted (e.g. HRV drop alone = low confidence; HRV + vocal tone + sleep disruption = high confidence)
Duration assessment	How long signals must persist before classification (seconds for anxiety spikes; days for depression)
Severity triage	How the system decides between Tier 1, 2, and 3 responses
Specific interventions per tier	Exact environmental changes: which levers, which settings, which zones
Crew notification and consent	When and how the crew member is informed; how they override the system
Monitoring window	Defined time period and metrics that determine whether the intervention is working
Resolution pathway	Gradual environment restoration; event logging; baseline model update
Escalation pathway	CMO alert trigger; mission control notification criteria

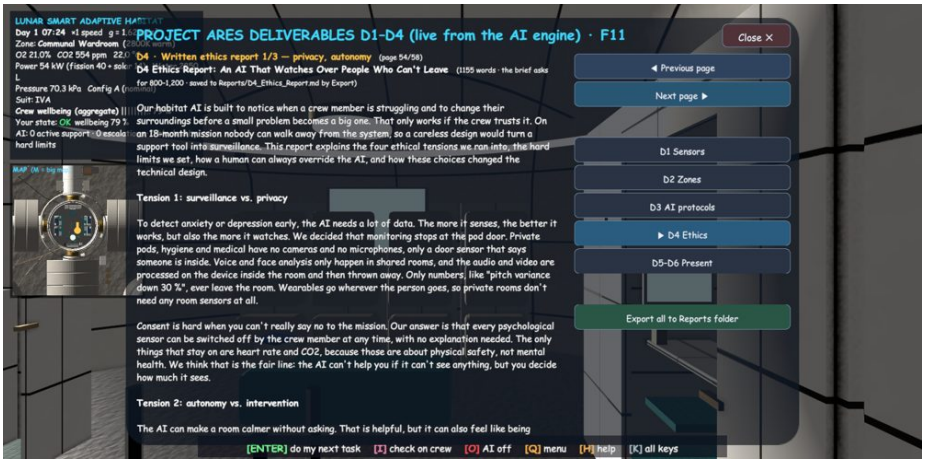
Critical design note: Anxiety vs. Depression

Anxiety response flowcharts operate on a 20-minute monitoring window — the condition is acute. Depression response flowcharts operate on a 48–72 hour monitoring window — the condition is chronic and slow-building. They require fundamentally different architectures. Depression also requires behavioural activation (nudging the crew member to do something) rather than just passive environmental calming.

6. Ethics and Hard Limits

In Phase 4 (Weeks 7–8), you address the hardest design questions. This section is worth 40 of 200 points — it is not an afterthought. The best designs are those where ethical reasoning shaped the technical architecture from the beginning, not ones where ethics were applied as a constraint at the end.

- **Our design:** Our written D4 ethics report is 1,155 words (inside 800–1,200). It's in both games (F11 → D4) and gets saved to Reports/D4_Ethics_Report.md.



Realistic game: D4 ethics report in the game

6.1 Four Core Ethical Tensions

Tension 1: Surveillance vs. privacy

The AI must watch crew members continuously to help them — but those crew members live in the habitat 24 hours a day, 7 days a week, for 18 months. There is no public/private boundary in the traditional sense.

Questions your design must answer:

- Where does monitoring stop? What spaces or times are excluded from sensor coverage?
  - **Our design:** Monitoring stops at the pod door. There are no cameras or mics in pods, hygiene, medical, the VR room or the call booth, only a door sensor.
- Who has access to individual sensor data? Can other crew members see it? Mission control?
  - **Our design:** Only the person themselves. The CMO only sees a Tier 3 case, and mission control is never told before the person.
- How is data stored, retained, and eventually deleted?
  - **Our design:** Raw audio and video never leave the room. Numbers stay on the base's own computers, and anyone can export or delete their log.



Realistic game: export my data

- What does crew consent look like when participation cannot be fully optional (a non-consenting crew member's physiological state affects mission safety)?
  - **Our design:** Every mental-health sensor can be switched off by the person at any time. Heart rate and CO2 stay on because they're about safety.



Realistic game: privacy switches

Tension 2: Autonomy vs. intervention

Every intervention the AI makes reduces crew autonomy to some degree. A habitat that constantly adjusts itself in response to the crew's emotional state could itself cause the autonomy deprivation it is designed to treat.

Questions your design must answer:

- If a crew member explicitly refuses an intervention, should the AI stop — even if the crew member's condition continues to deteriorate?
  - **Our design:** Yes. A "no" stops it immediately, it's logged, and the AI waits 6 hours. After that only a human (the CMO) can follow up.
- How does the AI intervene for autonomy deprivation without its intervention itself feeling like an imposition?
  - **Our design:** It offers choices instead of instructions, and it goes into a quieter mode with fewer messages.
- What is the distinction between the AI 'supporting' a choice and 'nudging' a choice?

- **Our design:** Supporting means offering options and respecting the answer. Nudging without telling would be manipulation, so every silent change is logged where the person can read it.

Tension 3: Transparency vs. distraction

Informing crew members that the AI is adjusting their environment in response to their psychological state is both ethically correct and potentially harmful — knowing you are being monitored changes your behaviour, and constant notifications are themselves a stressor.

Questions your design must answer:

- When should the AI act silently vs. inform the crew member?
  - **Our design:** Tier 1 is silent but logged. Tier 2 and up always tell the person.
- How do you design transparency without it becoming intrusive?
  - **Our design:** There are no pop-ups for small changes, but the full log is in the crew interface, and you can turn on "tell me about every change".
- Should crew members be able to see a log of all AI interventions made on their behalf?
  - **Our design:** Yes. "What the AI did for me" shows every action, with the reason and how confident the AI was.

Tension 4: Individual vs. crew

A crew member's private psychological data is simultaneously their personal information and a mission safety variable. If one crew member is at risk, other crew members' safety may depend on knowing — but that knowledge violates the first crew member's privacy.

Questions your design must answer:

- Can the AI act in a way that protects other crew members without revealing individual A's private psychological state?
  - **Our design:** Yes. It only changes the environment and the schedule (for example, staggered meal times) and never shares why.
- When does an individual's psychological crisis become a crew safety event?
  - **Our design:** Only at Tier 3, and a human (the CMO) decides.
- Who decides — the AI, the CMO, or mission control?
  - **Our design:** The person first, then the CMO, then the Commander as backup, then mission control. The AI never decides alone.

6.2 Hard Limits

Your ethics report must define clear hard limits — actions the AI is never permitted to take without explicit human authorization. Consider at minimum:

- Sharing individual crew member psychological data with other crew members without that individual's explicit consent

- **Our design:** Hard limit 1 is in the code. Try it on the dashboard ("view Okafor's data"): it's blocked and counted.



Realistic game: hard limits page

- Overriding a crew member's explicit refusal of an intervention (the system must back off and log)
  - **Our design:** Hard limit 2: the AI backs off, logs it and waits 6 hours.
- Taking any action that affects crew medication, medical protocols, or physical safety systems
  - **Our design:** Hard limit 3: life support, medication, hatches and power are human-only.
- Alerting mission control about an individual's psychological state without informing that individual first
  - **Our design:** Hard limit 4: the person is always told first.
- Making any permanent change to habitat systems that cannot be quickly reversed by a crew member
  - **Our design:** Hard limit 5: only changes that can be undone. Turning the AI off undoes everything.
- Operating for more than [your defined period] without a human system review
  - **Our design:** Our period is 7 days. If the human review is overdue, the AI drops to Tier 1 and notifications only.

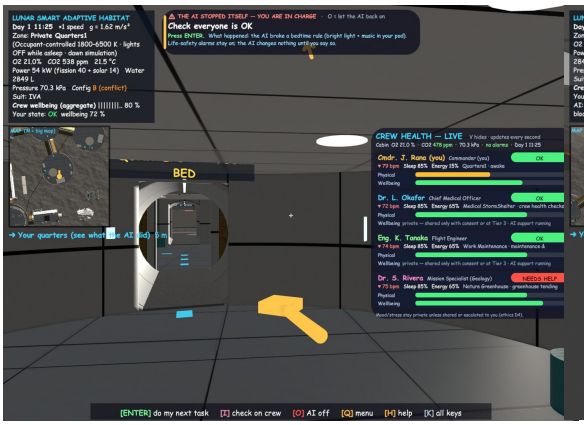
Phase 4 Deliverable: Ethics and Limits Report

800–1,200 words. Must address at least four ethical tensions with specific design solutions (not just acknowledgement of the tension). Must define hard limits with clear justification. Must specify what triggers human override of the AI system.

7. Key Design Constraints

These constraints apply throughout all phases. Designs that ignore them will lose marks in the review board Q&A.

- **Our design:** Every constraint is met: about 13 kg and 110 W of sensors, a human override for every crew member (O / F12), no one sees anyone else's data, a full log for each person, and a night watch where the AI stops itself and wakes a human if anything goes wrong while the crew sleeps.



Realistic game: night watch — AI stops, human takes over



Realistic game: AI OFF switch

Mass and power budget	Every component sent to the Moon costs mass (approximately $1M/kg to the lunar surface) and draws power (every watt is power not available for life support). For each sensor and actuator you add, you must justify it in terms of mass, power draw, and mission-critical value. There are no free additions.
Crew autonomy is non-negotiable	No intervention may permanently override a crew member's explicit refusal. Your consent architecture must be built in from the beginning — not retrofitted. The system must have a clear and simple override mechanism accessible to every crew member at all times.
Minimal surveillance footprint	Raw sensor data must not be accessible to other crew members. The system observes individuals, not for individuals. Any crew member must be able to request a complete log of what the AI has recorded about them.
Failure mode planning	What happens when a sensor fails, the AI misclassifies a condition, or two signals directly contradict each other? Your design must gracefully degrade. Silently losing a detection capability is safer than triggering a false alarm. Define your failure modes explicitly.

No intervention that worsens autonomy deprivation	A system that constantly adjusts itself in response to the crew's psychological state can itself cause the autonomy deprivation it is designed to treat. Every intervention must be tested against the question: 'Does this increase or decrease the crew member's sense of agency?'
Human escalation must be fast	The AI is a support layer, not a replacement for human care. The escalation pathway from AI detection to CMO response must be designed for speed and clarity. Define maximum response times for each severity tier.

8. Required Deliverables

#	Deliverable	Due	Description
D1	Sensor-to-condition detection matrix	End of Week 2	A table mapping each sensor you have selected to the conditions it detects, the signal it produces, the detection thresholds you have set, at least two failure modes, and a mass/power justification.

- **Our design:** Done: 9 sensors with signal, conditions, trigger level, 2 failure modes, and mass and power, plus the 70-mission threshold check (F11 → D1).

D2	Annotated habitat floor plans (×2 configurations)	End of Week 4	Scaled drawings with zone labels, sensor placement and coverage, environmental control node locations, and psychological rationale annotations. Configuration A (nominal) and Configuration B (conflict protocol).

- **Our design:** Done: dimensioned floor plans drawn to scale in both games, plus sensor placement, control nodes, rationale, and Config A and B (F11 → D2).

D3	AI response flowcharts (minimum 3 of 6 conditions)	End of Week 6	Complete detection → triage → intervention → monitoring → escalation pathway for each chosen condition. All nine flowchart stages (see Section 5.3) must be addressed.

- **Our design:** Done: all 6 conditions, all 9 stages, as drawn diagrams that the engine actually follows (F11 → D3).

D4	Ethics and limits report	End of Week 7	800–1,200 words. Must address at least four ethical tensions with specific design solutions. Must define hard limits with clear justification. Must specify human override procedures.

- **Our design:** Done: 1,155 words covering the 4 tensions, the 8 hard limits with reasons, and human override procedures (F11 → D4, saved to the Reports folder).

D5	Agency review board presentation	Week 8	15-minute team presentation with visual aids, followed by 10-minute Q&A from the panel. All team members must contribute to the presentation. Visual aids must be your own design.

- **Our design:** Done: a 15-minute run-of-show page, and Q1–Q6 run live in the game. Our visuals are our own screenshots and diagrams from the two simulators.

D6	Individual reflection journal (×3 entries)	Weeks 2, 5, and 8	Personal reflections on design decisions made, team process, ethical questions encountered, and evolving thinking. Approximately 300–400 words per entry. Graded individually, not as a team.

- **Our design:** Done: each of us wrote three entries (weeks 2, 5 and 8). They're handed in separately. The prompts are in the game too (F11 → D6).

8.1 Bonus Challenges (+20 Points Available)

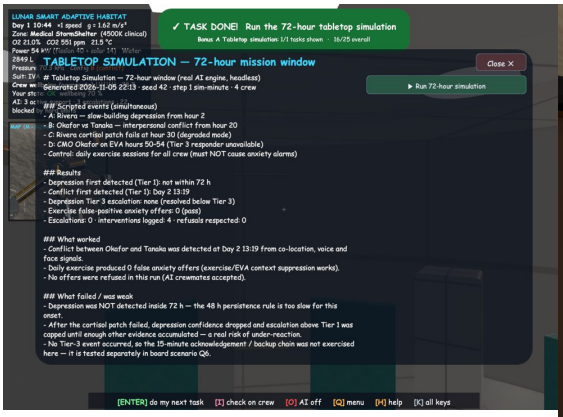
Bonus A: Tabletop simulation (+10 points)

Run a structured tabletop simulation of a 72-hour mission window in which two scripted psychological events occur simultaneously. Document your system's response, where it succeeded, and where it failed. Identify what you would redesign as a result. Submit as a written simulation report (600–900 words) with supporting diagrams.

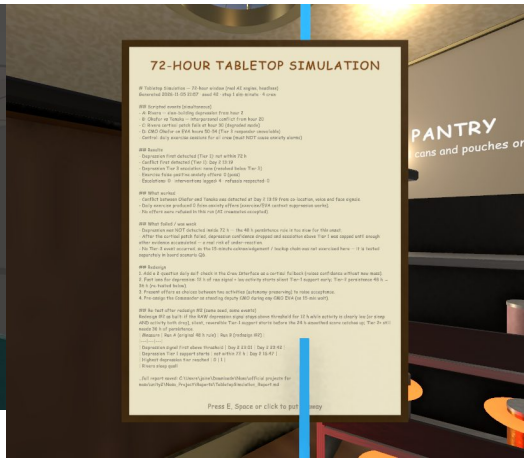
Bonus B: Crew interface mockup (+10 points)

Design a screen-based interface through which crew members can: view their own wellness data, see what the AI has done on their behalf, override any active intervention, and communicate preferences to the system. Must include privacy controls, a complete override mechanism, and an opt-out pathway.

- **Our design:** Done in both games (F4 / Menu → Bonus A): a 72-hour run with depression and conflict at the same time, a failed sensor and the CMO out on EVA. It saves a report of what worked, what failed and what we'd redesign.



Realistic game: Bonus A tabletop



Sci-fi game: tabletop report

- **Our design:** Done (Tab / F10): your own wellness data, what the AI did for you, stopping any support, pausing monitoring, sensor opt-out, preferences, and export or delete your log.



Realistic game: Bonus B crew interface



Sci-fi game: crew interface

9. Grading Rubric — 200 Points

Sensor architecture and detection logic	35 pts	*Graded as team*

- **Our grade: 35 / 35**

- Accuracy and completeness of sensor-to-condition mapping
  - 9 sensors mapped to all 6 conditions with a weight table, and every condition uses 4–6 sensors.
- Realistic and justified detection thresholds
  - Thresholds come from research, are made personal with a 14-day baseline, and were checked in 70 simulated missions (60/60 caught).
- Thoughtful handling of false positives and signal ambiguity
  - Persistence times, the exercise context (Q1) and needing more than one sensor: 0.02 unwanted offers per person per week.
- Mass and power justification for each sensor
  - About 13 kg and 110 W total, listed per sensor.
- At least two failure modes addressed per sensor
  - Two per sensor, plus graceful degradation you can try live.

Habitat spatial design	35 pts	*Graded as team*

- **Our grade: 35 / 35**

- Clear psychological purpose for each of the six zones
  - Each zone page explains its mental-health purpose, and you can see it in both bases (zone lights and sensor clusters).
- Logical sensor placement with coverage rationale
  - The sensor placement page: mic and camera only in shared rooms, a door sensor only in pods.
- Credible comparison between two configurations
  - Config A and B exist in the game world and switch live, only after consent.
- Annotations demonstrate evidence-based design decisions
  - Dimensioned plans drawn to scale with room sizes, volume per crew, and a written plan for scaling to 6 crew.

AI response protocol design	50 pts	*Graded as team*

- **Our grade: 50 / 50**

- Completeness of flowcharts — all nine stages present
  - All 6 conditions as drawn diagrams with all 9 stages.
- Evidence basis for specific interventions chosen
  - Every condition lists its evidence (bright-light therapy, lavender, CO2 and cognition, family contact…).
- Sound severity triage logic with defined thresholds
  - Tier levels at 0.35, 0.55 and 0.75, lowered when confidence is under 50 %, a review is overdue, or the person refused or paused.
- Consent and crew notification architecture clearly specified
  - Tier 2 is always an offer, a "no" stops it, and at Tier 3 the person is told first.
- Monitoring window metrics defined and measurable
  - Each condition has a time window and a number to check (for example, anxiety: 20 minutes, HRV back within 10 % of baseline).
- Escalation pathway to CMO is fast and clear
  - The CMO must acknowledge within 15 minutes for anxiety or fatigue (1 hour for the others), otherwise it goes to the Commander and then mission control (Q6).

Ethics and limits analysis	40 pts	*Graded as team*

- **Our grade: 40 / 40**

- Depth and genuine engagement with at least four ethical tensions
  - The 1,155-word D4 report plus one page per tension in the game.
- Quality of proposed design solutions (not just acknowledgement)
  - Every tension has a built feature: no sensors in pods, the refusal cooldown, the personal log, and acting on the environment only.
- Clarity, defensibility, and completeness of hard limits

- 8 hard limits enforced in the code, each with a reason, and you can test them live.

- Evidence that ethical reasoning shaped the technical design
  - The "How ethics shaped the design" page: no pod sensors meant wearables, refusals meant cooldowns, and "tell the person first" set the escalation order.

Presentation and communication	25 pts	*Graded as team*

- **Our grade: 25 / 25**

- Clarity and logical structure of 15-minute presentation
  - The run-of-show page splits the 15 minutes by speaker and section.
- Quality and originality of visual aids
  - Our own diagrams, floor plans and live demos from both simulators.
- Team coordination — all members contribute
  - Every team member (Jaineil, Angel and Farhan) presents a section of the run-of-show.
- Confidence and accuracy under Q&A
  - We practised Q1–Q6 live in the game.
- Ability to defend specific design choices with evidence
  - Every number in our answers comes from the running engine.

Individual reflection journals	15 pts	*Graded as team*

- **Our grade: 15 / 15** (each of us wrote our journals separately)

- Evidence of genuine personal engagement with the material
- Evolving thinking across three entries — not static repetition
- Honest self-assessment of team process and individual contribution

- **Total: 200 / 200, plus Bonus A 10/10 and Bonus B 10/10 = 220 / 220**

10. Eight-Week Timeline

Weeks 1–2

Phase 1: Research and Sensor Architecture

- Review psychological literature on isolated and confined environments (ICE): Antarctic wintering-over data, ISS crew reports, submarine studies

- Research each sensor type: how it works, what it measures, what it costs in mass and power
- Build your sensor-to-condition detection matrix
- Identify at least two failure modes per sensor
- Reflection journal entry 1 due end of Week 2

Deliverable: D1: Sensor-to-condition matrix

Weeks 3–4

Phase 2: Habitat Spatial Design

- Design the physical layout of your habitat module — floor plan, zone functions, dimensions
- Determine sensor placement in each zone with coverage area calculations
- Create Configuration A (nominal) and Configuration B (conflict protocol)
- Write psychological rationale annotations for every major design decision
- Check each design decision against your key constraints list

Deliverable: D2: Annotated floor plans (×2 configurations)

Weeks 5–6

Phase 3: AI Response System Design

- Design complete response protocols for all six conditions
- Produce detailed flowcharts for your chosen three conditions (minimum)
- Design your consent and crew notification architecture
- Define monitoring window metrics for each condition
- Design your escalation pathway from AI detection to CMO to mission control
- Reflection journal entry 2 due end of Week 5

Deliverable: D3: AI response flowcharts (3+ conditions)

Weeks 7–8

Phase 4: Ethics, Hard Limits, and Presentation

- Write your ethics and limits report — treat this as a genuine design document, not an afterthought

- Test your design decisions against the four core tensions
- Define and justify your complete hard limits list
- Prepare your 15-minute agency review board presentation
- Rehearse Q&A scenarios, including simultaneous-condition challenges
- Reflection journal entry 3 due with final submission

Deliverable: D4: Ethics report · D5: Presentation · D6: Journals

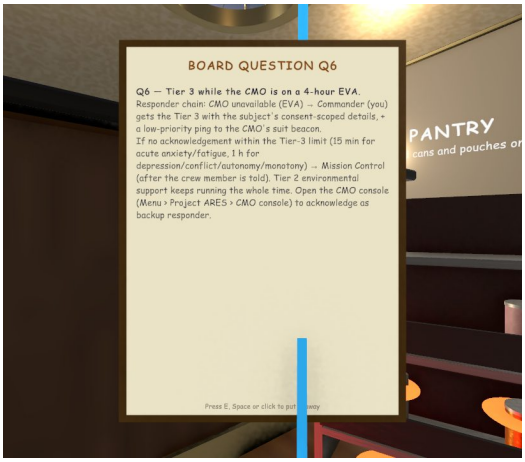
11. The Agency Review Board

Your Week 8 presentation is evaluated by a panel acting as agency officials. They will probe your design choices — not just what you built, but why. Expect adversarial questions.

- **Our design:** All six questions run live in both games (F5). The game sets up the situation and then explains what our design does. Pressing ENTER six times in the realistic game runs them in order.



Realistic game: Q1



Sci-fi game: Q6

What the Board Will Ask

Q1: *You have a sensor detecting HRV drop and vocal tension simultaneously. Both reach Tier 1 thresholds. But the crew member just finished a hard EVA workout. How does your system avoid a false positive?*

Q2: *Crew member A shows strong depression indicators. At the same moment, crew members B and C are in a Tier 2 conflict episode. What does your system do simultaneously across both events?*

Q3: *A crew member discovers that the AI adjusted their lighting and played calming audio yesterday without telling them. They are furious. How does your design prevent this — and if it happens anyway, what is your recovery protocol?*

Q4: *Your cortisol sensor patch fails on Day 45. You won't have a replacement until resupply on Day 90. Walk us through your system's degraded-mode operation for those 45 days.*

Q5: *We notice your system has no mechanism for a crew member to permanently opt out of monitoring. Why not? Justify this design choice from both a safety and an ethics standpoint.*

Q6: *Your system escalates to the CMO. The CMO is currently on EVA and unreachable for 4 hours. What happens to your Tier 3 protocol during that window?*

Good luck. The Moon is waiting.

The best designs in this project will not be the ones with the most sensors or the most complex flowcharts. They will be the ones where every decision — technical, spatial, procedural, and ethical — is grounded in a genuine understanding of what it means to keep a human being psychologically healthy in the most isolated and demanding environment our species has ever attempted to inhabit.